

Upwind Aronson–Bénilan counterexample

Siavash Sadeghi

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Abstract

The uniform estimate fails with fixed data bounds and $G(p)=1-p$. The proof selects γ after the mesh; the numerical experiments are corroboration, not a uniform choice of γ .

Provenance and review. Separated and polished from the supplied AI-assisted research note prepared with ChatGPT. Codex reviewed the mathematical argument and accompanying checks. This is a research claim awaiting independent mathematical review; no independent human audit or formal proof verification is claimed.

1 Entry 505: a counterexample to the uniform upwind estimate

The exact scheme and initial-data assumptions used here are those of catalogue entry 505 [1]; the originating scheme is David–Ruan [2]. Set

$$X = T = p_H = 1, \quad G(p) = 1 - p.$$

Thus $G(1) = 0$ and $G' = -1$. We prove that a single fixed initial-data bound C_0 cannot yield a constant independent of both h and γ in

$$w_i(t) := \frac{p_{i+1} - 2p_i + p_{i-1}}{h^2} + 1 - p_i \geq -\frac{C}{\gamma t}. \quad (1)$$

On the grid $i = -M, \dots, M$, $h = 1/M$, the exact density equation is

$$\dot{n}_i = \frac{F(n_i, n_{i+1}) - F(n_{i-1}, n_i)}{h^2} + n_i(1 - p_i), \quad F(a, b) = \max(a, b)(b^\gamma - a^\gamma), \quad p_i = n_i^\gamma.$$

The reflected ghosts are $n_{-M-1} = n_{-M+1}$ and $n_{M+1} = n_{M-1}$. The admissible initial data satisfy $0 \leq p_i(0) \leq 1$ and one fixed bound C_0 on each of

$$h \sum_i n_i(0), \quad h \sum_i p_i(0), \quad \sum_{i=-M}^{M-1} |n_{i+1}(0) - n_i(0)|, \quad h \sum_i |\dot{n}_i(0)|.$$

Unrestricted sums include both endpoints. The proof below constructs such data with $C_0 = 10$ along a sequence of even meshes and finite exponents.

1.1 A positive pressure profile on a fixed core

Let M be even, $M \geq 4$, $h = 1/M$, $K = M/2$, and

$$I_h = \{-K, \dots, K\}.$$

The full computational grid is $\{-M, \dots, M\}$. First consider a Dirichlet problem on the smaller core, with pressure zero at $\pm(K+1)$. Write

$$\eta_h = \text{arcosh}(1 + h^2/2), \quad P_i^h = 1 - \frac{\cosh(\eta_h i)}{\cosh(\eta_h (K+1))}, \quad i \in I_h.$$

Direct substitution gives

$$\Delta_h P_i^h + 1 - P_i^h = 0, \quad P_{\pm(K+1)}^h = 0. \quad (2)$$

This profile is positive, strictly less than one, symmetric, and strictly decreasing from the centre to the right endpoint. Moreover, with $c = \tanh(1/2) > 0$,

$$\frac{P_K^h}{h} \rightarrow c, \quad \frac{P_{K-1}^h}{P_K^h} \rightarrow 2, \quad \frac{P_{K-1}^h - P_K^h}{h} \rightarrow c. \quad (3)$$

Indeed, $\eta_h = h + O(h^3)$ and $(K+1)\eta_h \rightarrow 1/2$.

For a positive core pressure p , with zero Dirichlet neighbours outside the core, define

$$D_{\gamma,h}(p)_i = \Delta_h p_i + \frac{1}{h^2} \sum_{\substack{j=i\pm 1 \\ p_j > p_i}} \left[\left(\frac{p_j}{p_i} \right)^{1/\gamma} - 1 \right] (p_j - p_i). \quad (4)$$

This is precisely the upwind diffusion term in the density equation divided by $n_i = p_i^{1/\gamma}$. Lower-pressure neighbours contribute only to $\Delta_h p_i$.

For each fixed h , there exists a positive profile $P^{h,\gamma}$ for every sufficiently large γ such that

$$D_{\gamma,h}(P^{h,\gamma})_i + 1 - P_i^{h,\gamma} = 0, \quad P^{h,\gamma} \rightarrow P^h \quad (\gamma \rightarrow \infty). \quad (5)$$

Here and below, a limit with h fixed is a finite-dimensional limit. To justify existence, put $\nu = 1/\gamma$. The left side of (5) is continuously differentiable in (ν, p) near $(0, P^h)$. At a tie $p_j = p_i$, its additional term is of the form

$$(p_j - p_i)_+ \left[\exp\left(\nu \log \frac{p_j}{p_i}\right) - 1 \right],$$

which is C^1 and is quadratic to leading order in the difference of the pressures. At $\nu = 0$, the derivative with respect to p is the invertible Dirichlet matrix $\Delta_h - I$. The implicit-function theorem applies. By closeness to P^h and local uniqueness, $P^{h,\gamma}$ is symmetric, lies strictly between zero and one, and is strictly decreasing away from the centre for sufficiently large γ .

1.2 An exact solution on the pinned core

Fix

$$a = \frac{3}{4}, \quad q = \frac{1}{2}, \quad t_0 = -\log a > 0.$$

Take the full-grid initial data

$$n_i(0) = \begin{cases} a(P_i^{h,\gamma})^{1/\gamma}, & i \in I_h, \\ 0, & i \notin I_h. \end{cases} \quad (6)$$

If the exterior densities are artificially pinned to zero, the core has the exact solution

$$p_i^D(t) = \theta_\gamma(t) P_i^{h,\gamma}, \quad (7)$$

$$n_i^D(t) = \theta_\gamma(t)^{1/\gamma} (P_i^{h,\gamma})^{1/\gamma}, \quad (8)$$

$$\theta_\gamma(t) = \frac{1}{1 + (a^{-\gamma} - 1)e^{-\gamma t}}. \quad (9)$$

Indeed, $D_{\gamma,h}$ is homogeneous of degree one in the pressure, so (5) gives

$$D_{\gamma,h}(\theta P^{h,\gamma}) + 1 - \theta P^{h,\gamma} = 1 - \theta.$$

The density equation becomes $\dot{n}_i^D = (1 - \theta)n_i^D$, equivalent to $\theta' = \gamma\theta(1 - \theta)$. Its initial value is a^γ .

For fixed $t < t_0$, $\theta_\gamma(t) \rightarrow 0$, whereas for fixed $t > t_0$, $\theta_\gamma(t) \rightarrow 1$ and $\theta'_\gamma(t) \rightarrow 0$. The waiting time t_0 is independent of the mesh.

1.3 Control of the full, unpinned scheme

We now use the actual reflected Neumann ghost values at $\pm M$, as required by the target. The scalar face flux is

$$F(n, m) = \max(n, m)(m^\gamma - n^\gamma).$$

It is nondecreasing in its second argument. Thus the finite ODE system is cooperative. The maximum principle gives $0 \leq n_i \leq 1$, and comparison with the zero-extended pinned-core solution gives

$$n_i(t) \geq n_i^D(t), \quad i \in I_h. \quad (10)$$

The extension is a subsolution outside the core because all fluxes entering a zero-density cell are nonnegative.

Fix h for now and make γ so large that $\min_{i \in I_h} n_i^D(0) > q$. This is possible because that minimum tends to $a > q$. Since n^D is increasing in time, all core densities then exceed q .

Bootstrap on the condition that all exterior densities are at most q . Define the nonnegative discrepancy

$$D(t) = \sum_{i \in I_h} (n_i(t) - n_i^D(t)).$$

Internal face fluxes cancel when the density equations are summed. At each core boundary, the upwind density is the core density. As $n \geq n^D$ and $p \geq p^D$, the difference of the outgoing boundary terms is bounded above, in the discrepancy equation, by $np_{\text{out}} \leq q^\gamma$. Also,

$$[n - n^{\gamma+1}] - [n^D - (n^D)^{\gamma+1}] \leq n - n^D.$$

Consequently,

$$D'(t) \leq D(t) + \frac{2q^\gamma}{h^2}, \quad 0 \leq D(t) \leq \frac{2(e^t - 1)}{h^2} q^\gamma. \quad (11)$$

In particular, because $n \mapsto n^\gamma$ is γ -Lipschitz on $[0, 1]$,

$$0 \leq p_i(t) - p_i^D(t) \leq C_h \gamma q^\gamma \quad (i \in I_h, 0 \leq t \leq 1) \quad (12)$$

while the bootstrap condition holds. Here C_h is independent of γ ; its dependence on the fixed mesh is harmless at this stage.

To control the exterior, use weights $\omega_{\pm M} = 1/2$ and $\omega_i = 1$ at all other nodes, and set

$$E(t) = \sum_{i \notin I_h} \omega_i n_i(t).$$

These half-weights exactly cancel the factor two generated by the reflected boundary stencil. Hence

$$E'(t) \leq E(t) + \frac{p_K(t) + p_{-K}(t)}{h^2}, \quad E(0) = 0.$$

Equations (7) and (12) imply

$$E(t) \leq \frac{2P_K^{h,\gamma}}{h^2} \int_0^t e^{t-s} \theta_\gamma(s) ds + C_h \gamma q^\gamma. \quad (13)$$

Each exterior density is at most $2E(t)$.

Choose the observation time

$$t_h = t_0 + \log \left(1 + \frac{qh^2}{8P_K^h} \right). \quad (14)$$

For small h , it lies below $T = 1$, and $t_h \rightarrow t_0$. For fixed h , dominated convergence gives

$$\int_0^{t_h} e^{t_h-s} \theta_\gamma(s) ds \longrightarrow e^{t_h-t_0} - 1 = \frac{qh^2}{8P_K^h}.$$

Thus the right side of (13), at $t = t_h$, tends to $q/4$. The convolution in that estimate is nondecreasing in t . It follows that, for all sufficiently large γ , the estimate gives $2E(t) < q$ throughout the interval up to any putative first exit time from the bootstrap condition. Such a first exit is impossible. The estimates therefore hold on the entire interval $[0, t_h]$.

Besides $p(t_h) \rightarrow P^h$ on the core, we need the pressure derivative to converge to zero. This is not inferred merely from convergence of the pressures. On $[0, 1]$, each internal density flux has Lipschitz constant at most $C_h \gamma$. At a core/exterior face, where the exterior density is at most q , the flux discrepancy is bounded by

$$(\gamma + 1)|n_i - n_i^D| + q^\gamma.$$

Using (11) in the actual density equations therefore gives

$$|\dot{n}_i - \dot{n}_i^D| \leq C_h \gamma q^\gamma.$$

For $\gamma \geq 2$, $n \mapsto n^{\gamma-1}$ is $(\gamma - 1)$ -Lipschitz on $[0, 1]$, and $|\dot{n}_i^D| \leq 1$. Consequently,

$$|\dot{p}_i - \dot{p}_i^D| \leq C_h \gamma^2 q^\gamma. \quad (15)$$

At the fixed time $t_h > t_0$, $\dot{p}_i^D(t_h) = \theta'_\gamma(t_h) P_i^{h,\gamma} \rightarrow 0$. We have proved

$$p_i(t_h) \longrightarrow P_i^h, \quad \dot{p}_i(t_h) \longrightarrow 0, \quad i \in I_h. \quad (16)$$

1.4 The negative defect and uniform admissibility

At the right core endpoint K , the only higher-pressure neighbour is $K - 1$ when γ is sufficiently large. The exact pressure equation is

$$\frac{\dot{p}_K}{\gamma p_K} = w_K + \frac{1}{h^2} \left[\left(\frac{p_{K-1}}{p_K} \right)^{1/\gamma} - 1 \right] (p_{K-1} - p_K). \quad (17)$$

Multiply by γ and use (16). For each fixed h ,

$$\lim_{\gamma \rightarrow \infty} \gamma w_K(t_h) = -A_h, \quad A_h = \frac{P_{K-1}^h - P_K^h}{h^2} \log \frac{P_{K-1}^h}{P_K^h}. \quad (18)$$

Equation (3) yields

$$A_h \sim \frac{\tanh(1/2) \log 2}{h}. \quad (19)$$

In particular, $t_h A_h \rightarrow \infty$.

The initial data meet every stated uniform bound. Both $n_i(0)$ and $p_i(0)$ are at most one, so their h -weighted sums are at most three. Symmetry and monotonicity of the core profile give total variation at most $2a$. Inside the core, the exact initial derivative is

$$\dot{n}_i(0) = (1 - a^\gamma)n_i(0) \geq 0.$$

There are just two nonzero initial exterior derivatives, each at most $a^{\gamma+1}/h^2$. Thus

$$h \sum_i |\dot{n}_i(0)| \leq 3 + \frac{2a^{\gamma+1}}{h}. \quad (20)$$

Choose a sequence of even $M \rightarrow \infty$, and for each corresponding h choose a finite integer γ_h sufficiently large that all preceding estimates apply,

$$\gamma_h w_K(t_h) \leq -\frac{1}{2}A_h, \quad \frac{a^{\gamma_h+1}}{h} \leq 1.$$

All four initial bounds hold with the same $C_0 = 10$, while

$$-\gamma_h t_h w_K(t_h) \geq \frac{1}{2}t_h A_h \rightarrow \infty.$$

This contradicts any constant in (1) depending only on the permitted fixed parameters.

Theorem 1.1. *The uniform estimate asked for in entry 505 is false under its stated assumptions, including the reflected Neumann ghost values and all four uniform initial-data bounds.*

1.5 A numerical check, not used in the proof

The accompanying script solves the stationary core problem and integrates the full ODE with a stiff solver. For $\gamma = 500M$, it gives:

M	γ	$\gamma w_K(t_h)$	$-A_h$	$-\gamma t_h w_K(t_h)$
10	5000	-2.31735	-2.31723	0.69618
20	10000	-5.46771	-5.46736	1.60877
40	20000	-11.84682	-11.84617	3.44751
80	40000	-24.64579	-24.64478	7.13147

The proof selects γ_h by the rigorous fixed- h limit and does not assume that the particular numerical choice $500M$ works at every mesh size.

References

- [1] M. Colbrook, AIM catalogue, [entry 505: A discrete Aronson–Bénilan estimate for an upwind growth scheme](#). Repository statement checked 4 October 2026.
- [2] N. David and X. Ruan, *An asymptotic preserving scheme for a tumor growth model of porous medium type*, ESAIM: Mathematical Modelling and Numerical Analysis **56** (2022), 121–150. [doi:10.1051/m2an/2021080](#).